ROCSCIENCE

Case11_Water_Seismic_Surcharge
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case11_Water_Seismic_Surcharge

Project Summary

File Name:	Case11_Water_Seismic_Surcharge.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.825s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.005
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No
Seismic Load Coefficient (Horizontal):	0.2

Loading

1 Distributed Load present

Distributed Load 1	
Distribution:	Constant
Magnitude [kPa]:	10
Orientation:	Vertical

Materials

soil 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	0 kPa
Phi	38 °
Water Surface	Water Table
Hu Type	Custom
Hu	1
Specify alternate strength type above water surface	No

soil 2	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	5.3 kPa
Phi	23 °
Water Surface	Water Table
Hu Type	Custom
Hu	1
Specify alternate strength type above water surface	No

soil 3	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	7.2 kPa
Phi	20 °
Water Surface	Water Table
Hu Type	Custom
Hu	1
Specify alternate strength type above water surface	No

Global Minimums

Method: bishop simplified

FS	0.729855
Center:	34.183, 47.891
Radius:	26.065
Left Slip Surface Endpoint:	21.718, 25.000
Right Slip Surface Endpoint:	56.837, 35.000
Resisting Moment:	31516.2 kN-m
Driving Moment:	43181.4 kN-m
Total Slice Area:	180.278 m2
Surface Horizontal Width:	35.119 m
Surface Average Height:	5.13335 m

Method: spencer

FS	0.734161
Center:	34.758, 47.414
Radius:	25.290
Left Slip Surface Endpoint:	23.045, 25.000
Right Slip Surface Endpoint:	56.792, 35.000
Resisting Moment:	29643.5 kN-m
Driving Moment:	40377.4 kN-m
Resisting Horizontal Force:	1049.95 kN
Driving Horizontal Force:	1430.13 kN
Total Slice Area:	171.704 m2
Surface Horizontal Width:	33.7464 m
Surface Average Height:	5.08808 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	10035
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	9646
Number of Invalid Surfaces:	389

Error Codes

Error Code -108 reported for 248 surfaces
Error Code -111 reported for 81 surfaces
Error Code -112 reported for 60 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge
-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.729855

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.70414	2.5375	-27.6954	soil 3	7.2	20	14.566	10.6311	11.2396	1.81283	9.42675	3.59377	1.78094
2	0.70414	7.42867	-25.9605	soil 3	7.2	20	16.4742	12.0237	18.5605	5.30735	13.2531	10.5395	5.23218
3	0.70414	11.9601	-24.2508	soil 3	7.2	20	18.1452	13.2434	25.1488	8.54483	16.604	16.9747	8.42985
4	0.70414	16.1465	-22.5639	soil 3	7.2	20	19.6046	14.3085	31.0664	11.5358	19.5306	22.9203	11.3845
5	0.70414	20.0009	-20.8974	soil 3	7.2	20	20.8732	15.2344	36.3641	14.2897	22.0744	28.3945	14.1048
6	0.70414	23.5348	-19.2492	soil 3	7.2	20	21.9683	16.0337	41.0848	16.8144	24.2704	33.4135	16.5991
7	0.70414	26.758	-17.6174	soil 3	7.2	20	22.9045	16.7169	45.2649	19.1173	26.1476	37.9915	18.8742
8	0.70414	29.6793	-16.0003	soil 3	7.2	20	23.6939	17.2931	48.9351	21.2044	27.7307	42.1408	20.9364
9	0.70414	32.3064	-14.3961	soil 3	7.2	20	24.3472	17.7699	52.122	23.0813	29.0407	45.8725	22.7912
10	0.70414	34.6458	-12.8034	soil 3	7.2	20	24.8733	18.1539	54.8482	24.7528	30.0954	49.1956	24.4428
11	0.70414	36.7034	-11.2207	soil 3	7.2	20	25.2798	18.4506	57.1336	26.2229	30.9107	52.1186	25.8957
12	0.70414	38.6213	-9.64667	soil 3	7.2	20	25.6799	18.7426	59.2083	27.4951	31.7132	54.8433	27.3482
13	0.70414	43.561	-8.07994	soil 3	7.2	20	27.9335	20.3874	65.8245	29.5924	36.2321	61.8589	32.2665
14	0.70414	49.6339	-6.51927	soil 3	7.2	20	30.8871	22.5431	74.0137	31.8589	42.1548	70.484	38.6251
15	0.70414	55.4403	-4.96345	soil 3	7.2	20	33.662	24.5684	81.6544	33.9351	47.7193	78.731	44.7959
16	0.70414	60.9825	-3.41129	soil 3	7.2	20	36.2667	26.4694	88.7646	35.8224	52.9422	86.6028	50.7804
17	0.70414	66.262	-1.86164	soil 3	7.2	20	38.708	28.2512	95.3599	37.5221	57.8378	94.1017	56.5796
18	0.70414	71.2797	-0.313353	soil 3	7.2	20	40.9924	29.9185	101.453	39.0348	62.4185	101.229	62.1943
19	0.70414	76.0361	1.23471	soil 3	7.2	20	43.1251	31.4751	107.056	40.3607	66.6954	107.986	67.6248
20	0.70414	80.5311	2.78367	soil 3	7.2	20	45.111	32.9245	112.177	41.4999	70.6774	114.371	72.8708
21	0.70414	84.7638	4.33467	soil 3	7.2	20	46.954	34.2696	116.825	42.4517	74.3731	120.384	77.9321
22	0.70414	88.7329	5.88887	soil 3	7.2	20	48.6572	35.5127	121.004	43.2152	77.7886	126.022	82.8073
23	0.70414	92.4366	7.44743	soil 3	7.2	20	50.2239	36.6562	124.719	43.789	80.9301	131.284	87.4954
24	0.70414	95.8722	9.01155	soil 3	7.2	20	51.6562	37.7015	127.973	44.1713	83.8021	136.166	91.9943
25	0.70414	99.0366	10.5825	soil 3	7.2	20	52.9557	38.65	130.768	44.3598	86.4083	140.662	96.3019
26	0.70414	101.926	12.1615	soil 3	7.2	20	54.1243	39.5029	133.103	44.3517	88.7515	144.767	100.416
27	0.70414	104.535	13.75	soil 3	7.2	20	55.1626	40.2607	134.977	44.1438	90.8336	148.476	104.332
28	0.70414	106.86	15.3493	soil 3	7.2	20	56.0711	40.9238	136.388	43.7321	92.6556	151.779	108.047
29	0.70414	108.892	16.961	soil 3	7.2	20	56.8502	41.4924	137.33	43.1122	94.2178	154.669	111.556
30	0.70414	110.627	18.5867	soil 3	7.2	20	57.4992	41.9661	137.797	42.279	95.5185	157.133	114.854
31	0.70414	112.054	20.228	soil 3	7.2	20	58.0176	42.3444	137.785	41.2264	96.5585	159.163	117.937
32	0.70414	113.165	21.8869	soil 3	7.2	20	58.4039	42.6264	137.281	39.9477	97.3336	160.744	120.796
33	0.70414	113.948	23.5653	soil 3	7.2	20	58.6566	42.8108	136.275	38.4353	97.8396	161.859	123.424
34	0.70414	114.392	25.2654	soil 3	7.2	20	58.7733	42.896	134.754	36.6803	98.0742	162.493	125.813
35	0.70414	114.483	26.9898	soil 3	7.2	20	58.7513	42.8799	132.703	34.6728	98.0297	162.625	127.952
36	0.70414	114.204	28.741	soil 3	7.2	20	58.5868	42.7599	130.101	32.4011	97.6996	162.231	129.83
37	0.710966	114.632	30.5309	soil 2	5.3	23	62.3288	45.491	124.522	29.8385	94.6835	161.282	131.443
38	0.710966	113.531	32.3633	soil 2	5.3	23	61.728	45.0525	120.618	26.9668	93.6511	159.736	132.769
39	0.710966	111.982	34.2336	soil 2	5.3	23	60.9481	44.4833	116.089	23.7788	92.3099	157.561	133.783
40	0.710966	109.957	36.1466	soil 2	5.3	23	59.9818	43.778	110.902	20.2534	90.6484	154.716	134.463
41	0.710966	105.535	38.1074	soil 2	5.3	23	61.2364	44.6937	109.22	16.4148	92.8056	157.249	140.834
42	0.710966	97.5544	40.1225	soil 2	5.3	23	57.5668	42.0154	98.7624	12.2664	86.496	147.277	135.01
43	0.710966	88.9324	42.1993	soil 2	5.3	23	53.2887	38.893	86.8312	7.69098	79.1402	135.149	127.458
44	0.710966	79.6464	44.347	soil 2	5.3	23	48.9185	35.7034	74.2718	2.64582	71.626	122.088	119.442
45	0.639747	63.1341	46.4605	soil 2	5.3	23	43.7383	31.9226	62.7189	0	62.7189	108.746	108.746
46	0.639747	54.4176	48.5434	soil 2	5.3	23	37.736	27.5418	52.3985	0	52.3985	95.1165	95.1165
47	0.700686	48.7791	50.8248	soil 1	0	38	36.8631	26.9047	34.4364	0	34.4364	79.675	79.675
48	0.700686	36.4746	53.3337	soil 1	0	38	27.269	19.9024	25.474	0	25.474	62.1031	62.1031
49	0.700686	22.9476	56.0009	soil 1	0	38	17.703	12.9206	16.5376	0	16.5376	42.7842	42.7842
50	0.700686	7.92525	58.8679	soil 1	0	38	8.23555	6.01076	7.69343	0	7.69343	21.3284	21.3284

Global Minimum Query (spencer) - Safety Factor: 0.734161

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.683195	2.29122	-26.7238	soil 3	7.2	20	23.6264	17.3456	29.562	1.68706	27.8749	17.6668	15.9798
2	0.683195	6.70486	-25.0034	soil 3	7.2	20	26.2485	19.2706	38.1008	4.93709	33.1637	25.859	20.922
3	0.683195	10.7878	-23.3067	soil 3	7.2	20	28.1836	20.6913	45.0107	7.94362	37.0671	32.869	24.9254
4	0.683195	14.5531	-21.6314	soil 3	7.2	20	29.5852	21.7203	50.6105	10.7162	39.8943	38.8781	28.1619
5	0.683195	18.0119	-19.9754	soil 3	7.2	20	30.5655	22.44	55.1347	13.2632	41.8715	44.0246	30.7614
6	0.683195	21.1744	-18.3366	soil 3	7.2	20	31.2081	22.9118	58.7596	15.5919	43.1677	48.4164	32.8245
7	0.683195	24.0491	-16.7132	soil 3	7.2	20	31.5766	23.1823	61.6197	17.7087	43.911	52.1384	34.4297
8	0.683195	26.6438	-15.1035	soil 3	7.2	20	31.72	23.2876	63.8197	19.6193	44.2004	55.2589	35.6396
9	0.683195	28.9651	-13.5059	soil 3	7.2	20	31.6767	23.2558	65.4416	21.3286	44.113	57.8332	36.5046
10	0.683195	31.0187	-11.919	soil 3	7.2	20	31.477	23.1092	66.551	22.8408	43.7102	59.9068	37.066
11	0.683195	34.3409	-10.3413	soil 3	7.2	20	32.4366	23.8137	70.2349	24.589	45.6459	64.316	39.727
12	0.683195	40.3509	-8.77148	soil 3	7.2	20	35.5071	26.0679	78.897	27.058	51.839	73.4183	46.3603
13	0.683195	46.1795	-7.20829	soil 3	7.2	20	38.2814	28.1047	86.7746	29.3393	57.4353	81.9329	52.5936
14	0.683195	51.7562	-5.65048	soil 3	7.2	20	40.7272	29.9003	93.8037	31.4351	62.3686	89.7741	58.339
15	0.683195	57.0833	-4.09685	soil 3	7.2	20	42.8813	31.4818	100.061	33.3471	66.7137	96.9894	63.6423
16	0.683195	62.1625	-2.54624	soil 3	7.2	20	44.7746	32.8718	105.609	35.0766	70.5327	103.618	68.5416
17	0.683195	66.9949	-0.99749	soil 3	7.2	20	46.4334	34.0896	110.503	36.6244	73.8785	109.694	73.07
18	0.683195	71.5813	0.550528	soil 3	7.2	20	47.8798	35.1515	114.787	37.991	76.7962	115.247	77.2562
19	0.683195	75.9216	2.09895	soil 3	7.2	20	49.1331	36.0716	118.5	39.1764	79.324	120.301	81.1247
20	0.683195	80.0155	3.64891	soil 3	7.2	20	50.2094	36.8618	121.675	40.1803	81.4952	124.877	84.6971
21	0.683195	83.8619	5.20154	soil 3	7.2	20	51.1231	37.5326	124.34	41.002	83.3382	128.994	87.9921
22	0.683195	87.4592	6.75802	soil 3	7.2	20	51.8863	38.0929	126.518	41.6402	84.8777	132.666	91.0262
23	0.683195	90.8053	8.31954	soil 3	7.2	20	52.5099	38.5507	128.229	42.0935	86.1352	135.907	93.8139
24	0.683195	93.8974	9.8873	soil 3	7.2	20	53.0028	38.9126	129.489	42.3598	87.1297	138.728	96.3681
25	0.683195	96.7323	11.4626	soil 3	7.2	20	53.3737	39.1849	130.314	42.4366	87.8777	141.137	98.7004
26	0.683195	99.3058	13.0467	soil 3	7.2	20	53.6295	39.3727	130.715	42.321	88.3938	143.142	100.821
27	0.683195	101.613	14.6411	soil 3	7.2	20	53.7766	39.4807	130.7	42.0095	88.6905	144.749	102.739
28	0.683195	103.649	16.2471	soil 3	7.2	20	53.8208	39.5131	130.278	41.4979	88.7797	145.962	104.464
29	0.683195	105.407	17.8664	soil 3	7.2	20	53.767	39.4736	129.453	40.7816	88.6709	146.784	106.002
30	0.683195	106.879	19.5006	soil 3	7.2	20	53.6194	39.3653	128.228	39.8551	88.3733	147.217	107.361
31	0.683195	108.058	21.1514	soil 3	7.2	20	53.3822	39.1911	126.607	38.7123	87.8949	147.261	108.548
32	0.683195	108.933	22.8209	soil 3	7.2	20	53.0588	38.9537	124.589	37.3461	87.2424	146.915	109.569
33	0.683195	109.494	24.5112	soil 3	7.2	20	52.6523	38.6553	122.171	35.7484	86.4226	146.179	110.43
34	0.683195	109.728	26.2245	soil 3	7.2	20	52.1655	38.2979	119.351	33.9102	85.4409	145.047	111.137
35	0.683195	109.621	27.9635	soil 3	7.2	20	51.6011	37.8835	116.123	31.8208	84.3023	143.518	111.697
36	0.638091	101.976	29.6717	soil 2	5.3	23	54.9294	40.327	112.073	29.5548	82.5184	143.368	113.814
37	0.638091	101.265	31.35	soil 2	5.3	23	54.1043	39.7213	108.208	27.1168	81.0916	141.169	114.052
38	0.638091	100.233	33.0589	soil 2	5.3	23	53.2114	39.0657	103.972	24.4249	79.5471	138.606	114.181
39	0.638091	98.8596	34.8017	soil 2	5.3	23	52.2524	38.3617	99.3527	21.4643	77.8884	135.671	114.207
40	0.638091	97.0177	36.5822	soil 2	5.3	23	52.0636	38.2231	95.7798	18.2179	77.5619	134.421	116.203
41	0.638091	92.0974	38.4048	soil 2	5.3	23	52.2764	38.3793	92.6816	14.7516	77.93	134.122	119.371
42	0.638091	85.5868	40.2746	soil 2	5.3	23	48.9342	35.9256	83.1372	10.9877	72.1495	124.599	113.611
43	0.638091	78.6239	42.1978	soil 2	5.3	23	45.6813	33.5374	73.3904	6.86722	66.5232	114.808	107.941
44	0.638091	71.1666	44.1815	soil 2	5.3	23	42.5243	31.2197	63.42	2.35698	61.063	104.746	102.389
45	0.648101	64.0862	46.2512	soil 2	5.3	23	38.2484	28.0805	53.6675	0	53.6675	93.624	93.624
46	0.648101	55.1922	48.4192	soil 2	5.3	23	32.8275	24.1007	44.2916	0	44.2916	81.2911	81.2911
47	0.698884	48.705	50.7776	soil 1	0	38	33.217	24.3866	31.2133	0	31.2133	71.9089	71.9089
48	0.698884	36.4683	53.3563	soil 1	0	38	24.208	17.7726	22.7479	0	22.7479	55.2921	55.2921
49	0.698884	22.9784	56.1025	soil 1	0	38	15.5082	11.3855	14.5728	0	14.5728	37.6536	37.6536
50	0.698884	7.94533	59.0623	soil 1	0	38	7.7155	5.66442	7.25013	0	7.25013	20.1226	20.1226

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 0.729855

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	21.7181	25	0	0	0
2	22.4222	24.6304	13.8899	0	0
3	23.1264	24.2876	30.3524	0	0
4	23.8305	23.9704	48.6979	0	0
5	24.5346	23.6778	68.3446	0	0
6	25.2388	23.4089	88.7994	0	0
7	25.9429	23.163	109.643	0	0
8	26.6471	22.9394	130.52	0	0
9	27.3512	22.7375	151.127	0	0
10	28.0553	22.5568	171.208	0	0
11	28.7595	22.3968	190.547	0	0
12	29.4636	22.2571	208.965	0	0
13	30.1678	22.1374	226.386	0	0
14	30.8719	22.0374	243.897	0	0
15	31.5761	21.957	261.646	0	0
16	32.2802	21.8958	279.224	0	0
17	32.9843	21.8538	296.257	0	0
18	33.6885	21.8309	312.407	0	0
19	34.3926	21.8271	327.369	0	0
20	35.0968	21.8423	340.864	0	0
21	35.8009	21.8765	352.64	0	0
22	36.505	21.9299	362.471	0	0
23	37.2092	22.0025	370.153	0	0
24	37.9133	22.0946	375.505	0	0
25	38.6175	22.2062	378.366	0	0
26	39.3216	22.3378	378.595	0	0
27	40.0257	22.4895	376.074	0	0
28	40.7299	22.6618	370.702	0	0
29	41.434	22.8551	362.399	0	0
30	42.1382	23.0699	351.107	0	0
31	42.8423	23.3066	336.788	0	0
32	43.5464	23.5661	319.427	0	0
33	44.2506	23.849	299.031	0	0
34	44.9547	24.1561	275.637	0	0
35	45.6589	24.4884	249.307	0	0
36	46.363	24.8471	220.137	0	0
37	47.0671	25.2332	188.256	0	0
38	47.7781	25.6525	157.372	0	0
39	48.4891	26.1031	124.15	0	0
40	49.2	26.5869	88.868	0	0
41	49.911	27.1062	51.8714	0	0
42	50.622	27.6638	13.3416	0	0
43	51.3329	28.263	-24.4695	0	0
44	52.0439	28.9076	-60.3945	0	0
45	52.7549	29.6026	-93.2041	0	0
46	53.3946	30.2758	-120.11	0	0
47	54.0344	31	-144.83	0	0
48	54.735	31.8599	-158.402	0	0
49	55.4357	32.8011	-170.59	0	0
50	56.1364	33.8399	-179.972	0	0
51	56.8371	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 0.734161

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	23.0453	25	0	0	0
2	23.7285	24.656	25.9485	10.3794	21.8014
3	24.4117	24.3374	54.7879	21.9152	21.8014
4	25.0949	24.0431	85.2487	34.0995	21.8014
5	25.7781	23.7721	116.384	46.5535	21.8014
6	26.4613	23.5238	147.481	58.9922	21.8013
7	27.1445	23.2974	178	71.1999	21.8014
8	27.8276	23.0923	207.533	83.0132	21.8014
9	28.5108	22.9079	235.773	94.3091	21.8014
10	29.194	22.7438	262.49	104.996	21.8014
11	29.8772	22.5996	287.517	115.007	21.8014
12	30.5604	22.4749	311.698	124.679	21.8014
13	31.2436	22.3695	336.349	134.54	21.8015
14	31.9268	22.2831	360.922	144.369	21.8014
15	32.61	22.2155	384.903	153.961	21.8014
16	33.2932	22.1665	407.855	163.142	21.8014
17	33.9764	22.1362	429.405	171.762	21.8014
18	34.6596	22.1243	449.233	179.693	21.8014
19	35.3428	22.1308	467.071	186.829	21.8015
20	36.026	22.1559	482.689	193.076	21.8015
21	36.7092	22.1994	495.893	198.357	21.8014
22	37.3924	22.2616	506.525	202.61	21.8014
23	38.0756	22.3426	514.451	205.781	21.8015
24	38.7588	22.4425	519.569	207.828	21.8014
25	39.442	22.5616	521.799	208.72	21.8014
26	40.1251	22.7001	521.083	208.433	21.8014
27	40.8083	22.8584	517.387	206.955	21.8014
28	41.4915	23.0369	510.698	204.279	21.8014
29	42.1747	23.236	501.021	200.408	21.8014
30	42.8579	23.4562	488.385	195.354	21.8014
31	43.5411	23.6982	472.838	189.135	21.8014
32	44.2243	23.9625	454.45	181.78	21.8014
33	44.9075	24.25	433.314	173.326	21.8015
34	45.5907	24.5615	409.545	163.818	21.8014
35	46.2739	24.898	383.287	153.315	21.8014
36	46.9571	25.2607	354.71	141.884	21.8014
37	47.5952	25.6243	328.832	131.533	21.8014
38	48.2333	26.013	301.246	120.498	21.8013
39	48.8714	26.4283	272.175	108.87	21.8014
40	49.5095	26.8718	241.881	96.7525	21.8014
41	50.1476	27.3454	210.539	84.2156	21.8014
42	50.7856	27.8512	178.796	71.5183	21.8014
43	51.4237	28.3919	148.142	59.2568	21.8014
44	52.0618	28.9704	119.282	47.7127	21.8014
45	52.6999	29.5905	93.0179	37.2072	21.8014
46	53.348	30.2676	68.8031	27.5212	21.8014
47	53.9961	30.9981	46.8144	18.7257	21.8013
48	54.695	31.8543	33.7016	13.4806	21.8014
49	55.3939	32.7938	22.0554	8.82214	21.8014
50	56.0928	33.834	13.2053	5.28213	21.8014
51	56.7917	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	20, 20 70, 20 70, 24 70, 31 70, 35 50, 35 30, 25 20, 25
Material Boundary	40, 27 52, 24 70, 24
Material Boundary	30, 25 40, 27 50, 29 54, 31 70, 31

Scenario-based Entities

Type	Coordinates (x,y)	Master Scenario
Water Table	20, 25 30, 25 50, 29 70, 33.3746	Assigned to: soil 1 soil 2 soil 3
Distributed Load	50, 35 70, 35	Constant DistributionOrientation: VerticalMagnitude: 10 kN/m2Creates Excess Pore Pressure: No